

Ryan Fan

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EDUCATION

National University of Singapore, School of Computing

Aug 2024 – May 2028

Bachelor of Science in Business Analytics | GPA: 4.81/5.0 (*First Class Honours*)

- Coursework: Software Engineering (Top Student), Data Structures and Algorithms (A), Database Systems (A+)
- Leadership: Chairperson, Sports Management Committee; Captain, Floorball (Raffles Hall)

TECHNICAL SKILLS

Languages: Java, Go, Python, SQL

Backend & Data: Spring Boot, PostgreSQL, Redis, Kafka, Airflow, Snowflake, dbt

AI & Agents: FastMCP, DSPy, ChromaDB, Pydantic

Infrastructure & Testing: Docker, AWS, GitHub Actions, Testcontainers, JUnit, pytest, Prometheus

EXPERIENCE

TikTok

Jan 2027 – Jul 2027

Incoming Software Engineer Intern | Social / Messaging

Centre for Strategic Infocomm Technologies (CSIT)

May 2026 – Jul 2026

Software Engineer Intern | Python, FastMCP, ChromaDB, DSPy

- Built an air-gapped agentic workflow for Linux telemetry analysis, coordinating triage, investigation, and IOC-enrichment agents to produce structured, evidence-backed security investigations.
- Designed a FastMCP retrieval layer combining ChromaDB semantic search with canonicalised IOC lookup and idempotent upserts, reducing redundant indicator processing by 75%.
- Developed a concurrent internet-to-air-gap threat-intelligence pipeline with bounded executors, retries, and isolated transactions, reducing end-to-end sync latency by 60%.
- Applied DSPy and typed Pydantic schemas to structure agent evidence and provenance, with signed, versioned delta exports reducing offline transfer bundle size by 4.1×

Secretlab

Jan 2026 – May 2026

Data Engineer Intern | Python, SQL, Airflow, Snowflake, dbt, AWS

- Provisioned Airflow on Astronomer across isolated staging and production environments, using AWS Secrets Manager and GitHub Actions CI/CD pipelines to deploy 20+ production DAGs.
- Migrated 150+ AWS EventBridge schedules into Airflow, centralising CloudWatch logging and retry visibility to reduce incident debugging time from 30 minutes to under 5 minutes.
- Reduced dashboard data latency by 2 hours by re-architecting dbt-to-Tableau refresh pipelines with upstream dependency resolution and Snowflake metadata checks.
- Built a FinOps cost-attribution platform ingesting AWS, Fivetran, and Snowflake spend data into a medallion architecture, achieving per-service cost visibility and identifying 16% cloud OPEX reduction opportunities.

PROJECTS

PayCore – Payment Gateway & Settlement Engine | Go, PostgreSQL, Redis, Kafka, Prometheus Jun 2026

- Built a Go payment-processing backend supporting authorisation, capture, and settlement workflows, using idempotency keys and optimistic locking to prevent duplicate balance updates under retries.
- Designed a claim-based settlement engine that selected eligible payments, computed per-merchant net positions, and used settlement status transitions to avoid duplicate settlement during retries.
- Implemented a transactional outbox so payment state changes committed in PostgreSQL could be published to Kafka reliably for downstream ledger and settlement consumers.

LedgerFlow – Event-Driven Transaction Ledger | Spring Boot, PostgreSQL, Kafka, Testcontainers May 2026

- Built a Spring Boot ledger service consuming PayCore Kafka events, using event deduplication and database transactions to record balance movements reliably.
- Implemented idempotent ledger writes with replay records and optimistic locking to handle retries and concurrent account updates.
- Validated retry, duplicate-event, and concurrent-transfer scenarios with 58 Testcontainers integration tests against PostgreSQL and Kafka.